Scaled Graphs

One picture or one square can stand for more than 1.

Each picture stands for 2

Fish we saw

Mon	🐟 🐟 🐟
Tue	🐟 🐟 🐟 🐟 🐟
Wed	🐟 🐟

Each 🐟 = 2

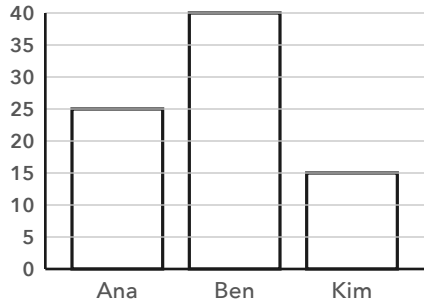
Monday has 3 fish pictures.

$3 \times 2 = 6$ fish

Count by 2s: 2, 4, 6.

A bar graph that counts by 5s

Books read



Each line is 5 more.
Ben read 40. Kim read 15.

$15 + 25 = 40$

Ben read 25 more than Kim.

Your turn!

Stars earned

Leo	☆ ☆ ☆ ☆

Each ☆ = 5

Leo has 20 stars

For teachers

Goal: read and draw picture and bar graphs where one picture or square stands for 2, 5 or 10 (3.MD.3), and answer how many more or less. Ask "What does one picture stand for?" before counting.

Scaled Graphs: Practice

Find what one picture or one square stands for. Then count by it.

Part 1 Each picture stands for 2 Count by 2s.

1)

Apples picked

Mia	
Sam	
Zoe	
Max	

Each  **= 2**

How many apples by Mia? **8**

How many more Sam than Zoe? **8**

Zoe and Mia together? **14**

How many in all? **38**

2)

Birds at the feeder

Mon	
Tue	
Wed	
Thu	

Each  **= 2**

How many birds on Mon? **12**

How many more Wed than Tue? **12**

Wed and Thu together? **26**

How many in all? **42**

Scaled Graphs: Practice (continued)

Part 2 Each picture stands for 5 or 10 Count by 5s or 10s.

3)

Stickers

Ava ☆☆☆☆☆

Leo ☆☆☆☆☆☆☆☆

Kim ☆☆☆

Jay ☆☆☆☆☆☆

Each ☆ = 5

How many stickers for Ava? **25**

How many more Leo than Kim? **25**

Jay and Ava together? **55**

How many in all? **110**

4)

Cupcakes sold

Fri 🧁🧁🧁🧁🧁🧁

Sat 🧁🧁🧁🧁🧁🧁🧁🧁🧁🧁

Sun 🧁🧁🧁🧁

Each 🧁 = 10

How many cupcakes on Fri? **60**

How many more Sat than Sun? **50**

Sun and Sat together? **130**

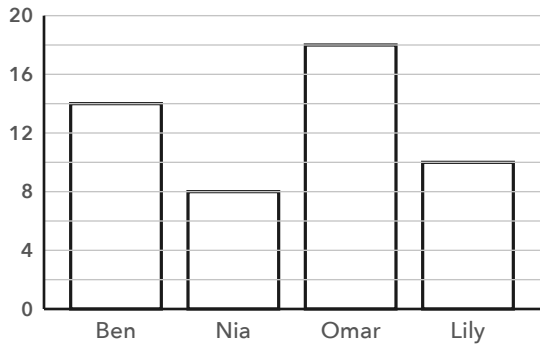
How many in all? **190**

Scaled Graphs: Practice (continued)

Part 3 Bar graphs Each line is 2 or 5 more.

5)

Laps run (by 2s)



How many laps by Ben? **14**

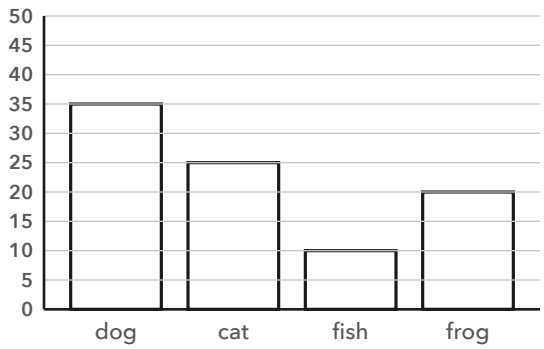
How many more Omar than Nia? **10**

Omar and Lily together? **28**

How many in all? **50**

6)

Votes for a pet (by 5s)



How many votes for frog? **20**

How many more dog than fish? **25**

Dog and fish together? **45**

How many in all? **90**

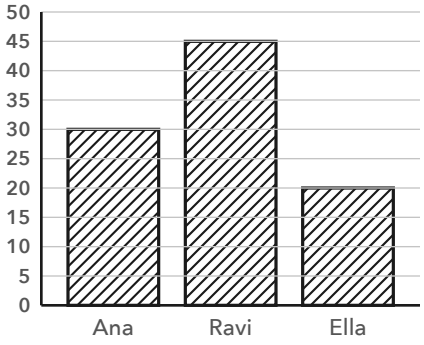
Scaled Graphs: Practice (continued)

Part 4 Make the graph Draw each bar to the right line.

7) **Pages read**

Ana	30
Ravi	45
Ella	20

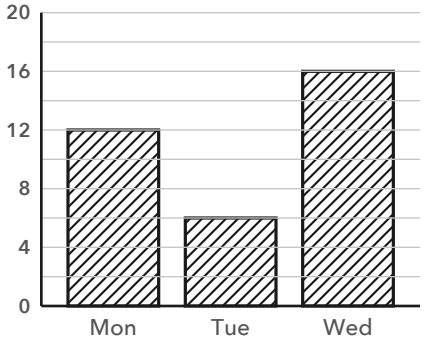
Make the bar graph (count by 5s)



8) **Shells found**

Mon	12
Tue	6
Wed	16

Make the bar graph (count by 2s)



Part 5 How many? No help this time.

9) each picture = 5: 7 pictures = 35

10) each picture = 2: 9 pictures = 18

11) each picture = 10: 6 pictures = 60

12) each picture = 5: 4 pictures = 20

13) each picture = 2: 8 pictures = 16

14) each picture = 10: 3 pictures = 30

15) each picture = 5: 9 pictures = 45

16) each picture = 2: 6 pictures = 12

Scaled Graphs: Practice (continued)

Part 5 How many? (continued)

17) each picture = 10: 8 pictures = 80

18) each picture = 5: 6 pictures = 30

Part 6 How many pictures? No help this time. How many groups?

19) each picture = 2: 7 pictures = 14

20) each picture = 5: 8 pictures = 40

21) each picture = 10: 4 pictures = 40

22) each picture = 2: 5 pictures = 10

23) each picture = 5: 3 pictures = 15

24) each picture = 10: 9 pictures = 90

Part 7 How many does it show?

25) each picture = 5: 2 pictures = 10

26) each picture = 2: 2 pictures = 4

27) each picture = 10: 2 pictures = 20

Scaled Graphs: Practice (continued)

Part 8 How many pictures?

28) each picture = 2: [3] pictures = 6

29) each picture = 10: [8] pictures = 80

30) each picture = 10: [5] pictures = 50

31) each picture = 5: [9] pictures = 45

32) each picture = 5: [6] pictures = 30

33) each picture = 5: [7] pictures = 35

34) each picture = 2: [9] pictures = 18

35) each picture = 10: [2] pictures = 20

36) each picture = 10: [6] pictures = 60

37) each picture = 5: [4] pictures = 20

38) each picture = 2: [2] pictures = 4

39) each picture = 2: [4] pictures = 8

Scaled Graphs: Practice (continued)

Part 8 How many pictures? (continued)

40) each picture = 10: [3] pictures = 30

41) each picture = 10: [7] pictures = 70

42) each picture = 5: [2] pictures = 10

43) each picture = 5: [5] pictures = 25

Part 9 Remember? Problems from earlier packets.

44) $9 \times 4 = 36$ so $9 \times 8 =$ [72]

45) 500 g + 350 g = [850] g

46) 6 cm by 5 cm: area = [30] sq cm

47) 15 minutes after 7:50 = [8:05]

48) $56 \div 7 =$ [8]

49) $700 - 365 =$ [335]

Scaled Graphs: Practice (continued)

Part 10 Word problems Draw a graph if it helps.

50) On a picture graph each star stands for 5. Kim has 6 stars. How many points is that?

Number sentence **6 × 5 = 30**

Answer **30** points

Draw or show your work

51) Each picture stands for 2 kids. The soccer row has 8 pictures. How many kids like soccer?

Number sentence **8 × 2 = 16**

Answer **16** kids

Draw or show your work

52) A bar graph counts by 10s. The red bar reaches 70 and the blue bar reaches 40. How many more is red?

Number sentence **40 + 30 = 70**

Answer **30** more

Draw or show your work

53) Each apple picture stands for 10 apples. A farm picked 90 apples. How many pictures?

Number sentence **90 ÷ 10 = 9**

Answer **9** pictures

Draw or show your work

54) Challenge: Each picture is 5 votes. Pizza has 7 pictures and tacos have 4. How many votes in all?

Number sentence **7 + 4 = 11, 11 × 5 = 55**

Answer **55** votes

Draw or show your work

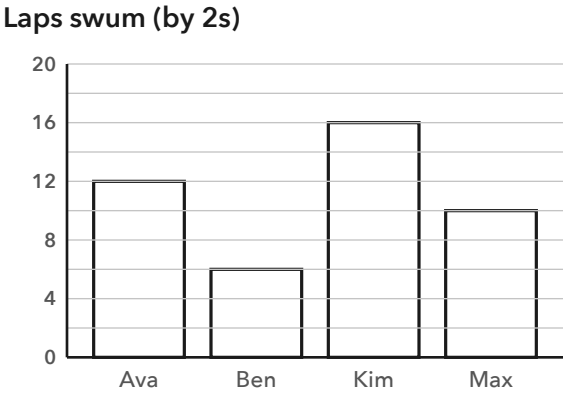
Check-In: Scaled Graphs

Show what you know. No help this time.

Mastery: 6 of 7 (80%)

Part 1 Read the graph

1)



How many laps by Max? **10**

How many more Kim than Ben? **10**

Ava and Max together? **22**

How many in all? **44**

Part 2 How many?

2) each picture = 10: 5 pictures = **50**

3) each picture = 2: 5 pictures = **10**

4) each picture = 2: 3 pictures = **6**

5) each picture = 5: 3 pictures = **15**

Part 3 How many pictures?

6) each picture = 2: **8** pictures = 16

7) each picture = 2: **6** pictures = 12